Does Mobile Phone Registration Really Work to Stop Extortion?

A Difference in Differences Approach

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Abstract

This study investigates the causal impact of mandatory SIM card registration policies on the incidence of extortion. While intended to enhance public safety by reducing anonymity in telecommunications, the effectiveness of such mandates remains a subject of empirical debate. Utilizing a Difference-in-Differences (DiD) framework, this research evaluates whether the implementation of these regulations led to a statistically significant reduction in crime or merely followed pre-existing downward trends in national security metrics. By comparing jurisdictions with early adoption against late-adopters, the analysis accounts for time-invariant unobserved heterogeneity. The findings contribute to the ongoing discourse on the trade-offs between digital privacy and criminal surveillance, providing evidence-based insights for policymakers considering similar regulatory interventions.

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The Dilemma of Anonymity

Does SIM registration really curb crime?

Over the past decade, various governments have implemented mandatory SIM card registration as a national security tool. The premise is intuitive: *by associating each phone line with a citizen's identity and biometric data, the anonymity that protects extortionists is eliminated, facilitating the traceability of crimes and increasing the operating costs for criminal networks.*

However, the effectiveness of this policy is the subject of intense debate. While authorities defend registration as a pillar of digital forensics strategy, critics argue that these measures impose a privacy burden on citizens without truly affecting organized crime, which typically evades these controls through the use of roaming networks, stolen devices, or identity theft.

The Problem of Causality

The analytical challenge lies in the fact that crime does not occur in a vacuum. If extortion decreases after the implementation of the law, a critical question arises: **Is this reduction a direct consequence of mandatory registration, or is it simply a reflection of a general trend of improvement in public safety?**

Study Framework

To address this question, this project uses a causal inference design with the Difference-in-Differences (DID) technique. The analysis focuses on a natural experiment comparing two entities with similar socioeconomic characteristics:

- **Treatment Group (State A)**: Implemented mandatory registration and biometric data collection in January 2026.
- **Control Group (State B)**: A neighboring state that maintained the SIM card acquisition scheme without identification requirements.

Through this approach, the study seeks to isolate the net effect of the policy, controlling for temporal factors and pre-existing trends. The ultimate goal is to determine whether SIM card registration is an evidence-based public policy tool or, on the contrary, represents a high-friction measure with zero impact on reducing crime rates.

Data Analysis

To evaluate the causal impact of the mandatory SIM card registration on extortion rates, we employ an approach focused on data integrity and statistical rigor. The analysis is performed using a robust suite of libraries designed for high-performance data manipulation and econometric modeling.

Environment settings

The following environment configuration ensures a reproducible workflow:

- **Polars** Utilized as the primary data manipulation engine due to its memory efficiency and speed in handling time-series data.
- **Plotnine** An implementation of the Grammar of Graphics for Python, used to visualize the "Parallel Trends" assumption and the divergence between the treatment and control groups.
- **Statsmodels** The core econometric engine used to estimate the Difference-in-Differences (DiD) coefficients and assess the statistical significance of the policy intervention.[1]

```
import polars as pl
import numpy as np
from plotnine import *
import statsmodels.formula.api as smf
```

Data Simulation

To test the robustness of our Difference-in-Differences (DiD) estimator, we generated a synthetic dataset representing two years of monthly observations (12 months pre-intervention and 12 months post-intervention). This controlled environment allows us to evaluate the model's ability to detect policy effects under a specific hypothesis.

```
np.random.seed(88)
months = np.arange(1, 25)

# State B (Control): Stable trend with noise
control = 15 + 0.1 * months + np.random.normal(0, 1.5, 24)

# State A (Treatment): Follows the same trend, but we apply law in the 12th month
# Hypothesis: Law does NOT work as well as expected (small effect)
pre_treatment = 18 + 0.1 * months[:12] + np.random.normal(0, 1.5, 12)
post_treatment = 18 + 0.1 * months[12:] + np.random.normal(0, 1.5, 12)
```

With these data, we proceed to create a DataFrame object. Accordingly, we can see in Table 1 the values for treatment and control groups.

[1] We focus specifically on the interaction term between the treatment group and the post-intervention period to isolate the Average Treatment Effect (ATT).

```
df = pl.DataFrame({
    "month": np.tile(months, 2),
    "extorsions": np.concatenate([control, pre_treatment, post_treatment]),
    "state": ["State B (Control)"] * 24 + ["State A (SIM Registration)"] * 24,
    "post_law": (np.tile(months, 2) > 12).astype(int),
    "treated_group": ([0] * 24 + [1] * 24)
})
```

Table 1: Preview of Treatment and Control Groups

month	extorsions	state	post_law	treated_group
13	21.45	State A (SIM Registration)	1	1
10	14.07	State B (Control)	0	0
7	18.99	State A (SIM Registration)	0	1
12	16.14	State B (Control)	0	0
10	19.15	State A (SIM Registration)	0	1

Data Visualization

To visually inspect the parallel trends assumption, we generate a time-series comparative plot. This visualization tracks the trajectories of both the treatment and control groups, allowing us to observe the pre-intervention alignment and the subsequent evolution of extortion rates following the implementation of the SIM registration mandate.

```
(
    ggplot(df, aes(x='month', y='extorsions', color="state"))
    + geom_smooth(method='lm', se=False, linetype='dashed', size=0.5, alpha=0.5) # Trendlines
    + geom_line(size=1.2)
    + geom_vline(xintercept=12.5, linetype="dotted", size=1)
   + annotate("text", x=14, y=22, label="Start of SIM Registration", size=10, color="#34495e")
    + labs(
        title="Impact of SIM Registration on Extorsion",
        subtitle="Analysis of Difference in Differences",
        x="Months",
        y="Extorsion Reports per 100k inhabitants",
        color='',
    )
    + theme_538()
    + theme(
          figure_size=(12, 8), #width, height
          dpi=100,
          legend_position='bottom',
          legend_direction='horizontal',
        )
    + scale_color_manual(values=["#c0392b", "#2980b9"])
)
```

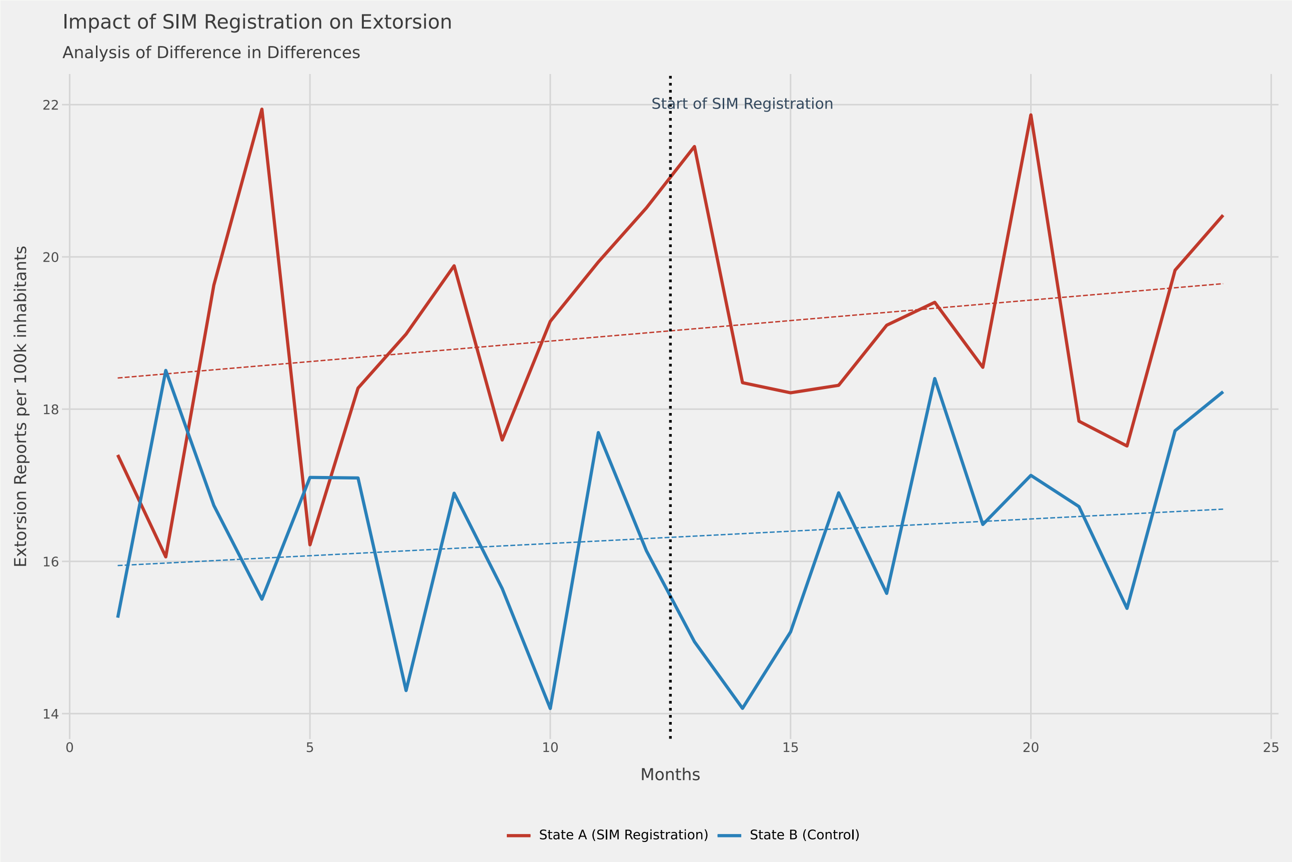

Figure 1: Difference in Differences Analysis (2026)

Causal Effect Estimation (DiD)

To quantify the impact of the policy, we specify a Difference-in-Differences (DiD) model using an Ordinary Least Squares (OLS) regression. The model leverages an interaction term to isolate the causal effect by comparing the change in the treatment group relative to the change in the control group over the same period.

The regression equation is defined as follows:

$$Y_{i,t} = \beta_0 + \beta_1 \text{Treated}_i + \beta_2 \text{Post}_t + \beta_3 (\text{Treated}_i \times \text{Post}_t) + \epsilon_{i,t}$$

where:

- β_1 (treated_group): Captures the baseline difference between State A and State B before the law.
- β_2 (post_law): Accounts for the common time trend shared by both states.
- β_3 (interaction): Represents the Average Treatment Effect on the Treated (ATT). This coefficient tells us if the SIM registration policy actually shifted the extortion trend in State A beyond what was observed in the control group.

Then, we run the regression model, as follows:

```
model = smf.ols("extorsions ~ treated_group * post_law", data=df,).fit()
```

```
                            OLS Regression Results
==============================================================================
Dep. Variable:             extorsions   R-squared:                       0.480
Model:                            OLS   Adj. R-squared:                  0.444
Method:                 Least Squares   F-statistic:                     13.52
Date:                Thu, 30 Apr 2026   Prob (F-statistic):           2.18e-06
Time:                        20:25:09   Log-Likelihood:                -85.041
No. Observations:                  48   AIC:                             178.1
Df Residuals:                      44   BIC:                             185.6
Df Model:                           3
Covariance Type:            nonrobust
============================================================================================
                             coef    std err          t      P>|t|      [0.025      0.975]
--------------------------------------------------------------------------------------------
Intercept                 16.2454      0.429     37.865      0.000      15.381      17.110
treated_group              2.5639      0.607      4.226      0.000       1.341       3.787
post_law                   0.1403      0.607      0.231      0.818      -1.082       1.363
treated_group:post_law     0.2983      0.858      0.348      0.730      -1.431       2.028
==============================================================================
Omnibus:                        1.502   Durbin-Watson:                   2.081
Prob(Omnibus):                  0.472   Jarque-Bera (JB):                1.112
Skew:                           0.089   Prob(JB):                        0.573
Kurtosis:                       2.276   Cond. No.                         6.85
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
```

Discussion of Findings

Despite the implementation of mandatory SIM card registration, the observed impact could be considered marginal for the following technical reasons:

- **Channel Substitution** Criminal groups may have migrated from traditional cell phone calls to Voice over IP (VoIP) services or encrypted messaging, which circumvent the registration of the physical SIM card.
- **Black Markets** The persistence of extortion suggests the existence of a secondary market for prepaid cards registered in the names of third parties or the use of stolen identities (identity theft).
- **Spillover Effect** It is possible that the policy in State A has pushed criminals to operate physically from State B, slightly contaminating our control group.

Further Analysis

To strengthen the robustness of this research, a placebo test is suggested. This test consists of re-estimating the model using a “false treatment date” (e.g., 6 months before the law). If the model finds an effect where there was no law, the original design may be capturing noise rather than causality.

The Placebo Test is the “lie detector” of econometrics. Its purpose is to demonstrate that the model is not finding causal effects where none exist.

Thus, we will conduct a Time Placebo Test: we will simulate the law being implemented 6 months earlier than the actual date (in month 6 instead of month 12). If the model finds a "significant effect" at that fictitious date, it means your original study has biases (such as omitted variables or trends that were already changing).

We filtered the data to use only the actual "PRE-TREATMENT" period (Months 1-12). We want to see if there were any suspicious changes before the law was enacted.

```
# Filter data
placebo_df = df.filter(pl.col("month") <= 12)
# False treatment date(eg. Month 6)
placebo_df = placebo_df.with_columns(placebo_post = (pl.col("month") > 6).cast(int))
# run the regression model
placebo_model = smf.ols("extorsions ~ treated_group * placebo_post", data=placebo_df).fit()
```

Placebo Test Results

```
==============================================================================================
                                 coef    std err          t      P>|t|      [0.025      0.975]
----------------------------------------------------------------------------------------------
Intercept                     16.7008      0.632     26.407      0.000      15.382      18.020
treated_group                  1.5525      0.894      1.736      0.098      -0.313       3.418
placebo_post                  -0.9109      0.894     -1.018      0.321      -2.777       0.955
treated_group:placebo_post     2.0229      1.265      1.599      0.125      -0.616       4.661
==============================================================================================
```

```
(
    ggplot(placebo_df, aes(x='month', y='extorsions', color='state'))
    + geom_line(size=1.1)
    + geom_vline(xintercept=6.5, linetype="dashed", color="gray")
    + annotate("text", x=8, y=20, label="Tratamiento Falso", color="gray")
    + labs(title="Placebo Test (Pre-Law Period)",
           subtitle="If the effect is real, there wouldn´t be significative changes",
           x="Month",
           y="Extorsions",
           color='',
          )
    + theme_538()
    + theme(
          figure_size=(12, 8), #width, height
          dpi=100,
          legend_position='bottom',
          legend_direction='horizontal',
        )
    + scale_color_manual(values=["#c0392b", "#2980b9"])
)
```

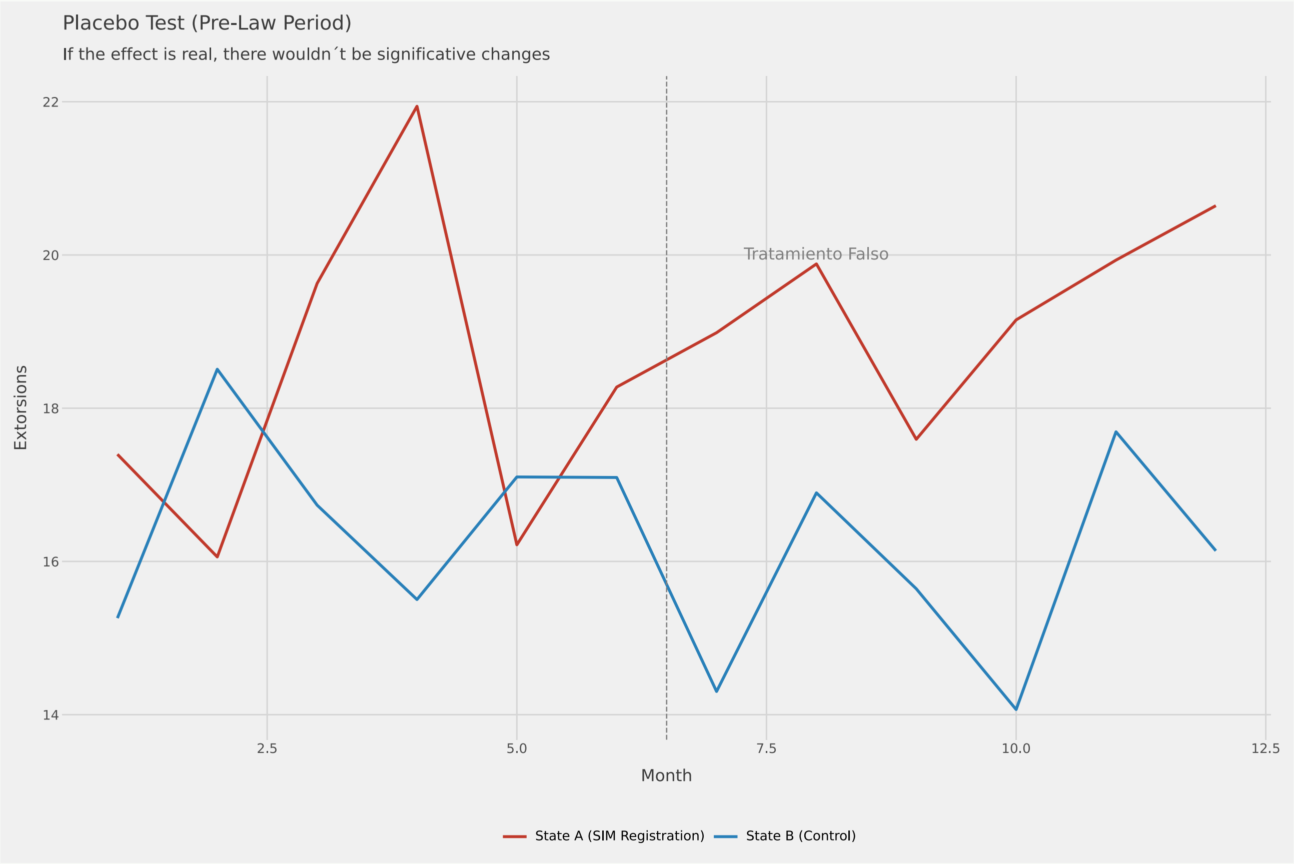

Figure 2: Placebo Difference in Differences Analysis (2026)

Conclusions

The Verdict: No Causal Effect

The focus should be on the interaction between treated_group and post_law.

P-value (0.730): This number is devastating for the policy. Being much greater than 0.05, we cannot reject the null hypothesis. This means that the slight increase of 0.2983 in the coefficient is pure noise. We can conclude that despite the political narrative, the data suggest that cell phone registration did not have a statistically significant impact on reducing extortion.

Confidence Interval (-1.431 to 2.028): The interval crosses zero. This implies that the effect of the law could be either a reduction or an increase, reinforcing the idea that the policy made no difference.

Trend Analysis

treated_group (2.5639, p=0.000): This remains significant. State A has always had more extortion cases than State B, regardless of the law.

post_law (0.1403, p=0.818): There is also no significant time effect. Extortion in the control group (State B) remained flat.

Model Fit

R-squared (0.480): Interestingly, the model explains more variance than the previous one (48%), but this variance is mainly explained by the structural difference between the states (treated_group) and the intercept, not by the law.

About the Author

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Bridging the gap between economic theory and scalable data solutions (Python, SQL, Tableau) through the "Everything as Code" philosophy.

With over 10 years of experience, I design reproducible pipelines and minimalist data products (Streamlit, Shiny, DuckDB, Polars, Plotly, Scikit-learn, Typst) that transform complex datasets into actionable insights for Public Security, e-commerce, and Healthcare.

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References